

CSC-335

HCI & Computer Graphics

Lec 5: User Needs

MileStone

1



Steps & Activities for M-01

<i>Step</i>	<i>Activities</i>	<i>Output</i>
Clarify the design problem and scope	• Select design challenge • Problem research • Identify existing offerings • Web searches: background research • Informal interviews	(See project brief)
Create project brief	• Define target audience • Problem definition • User needs • Activities • Overview of competitive landscape (preliminary) • System goals	Project brief
Plan needs finding study	• Establish study goals • Develop recruiting criteria • Design interview protocol	Study plan

Understanding User Needs



Vending Machine





Vending Machine



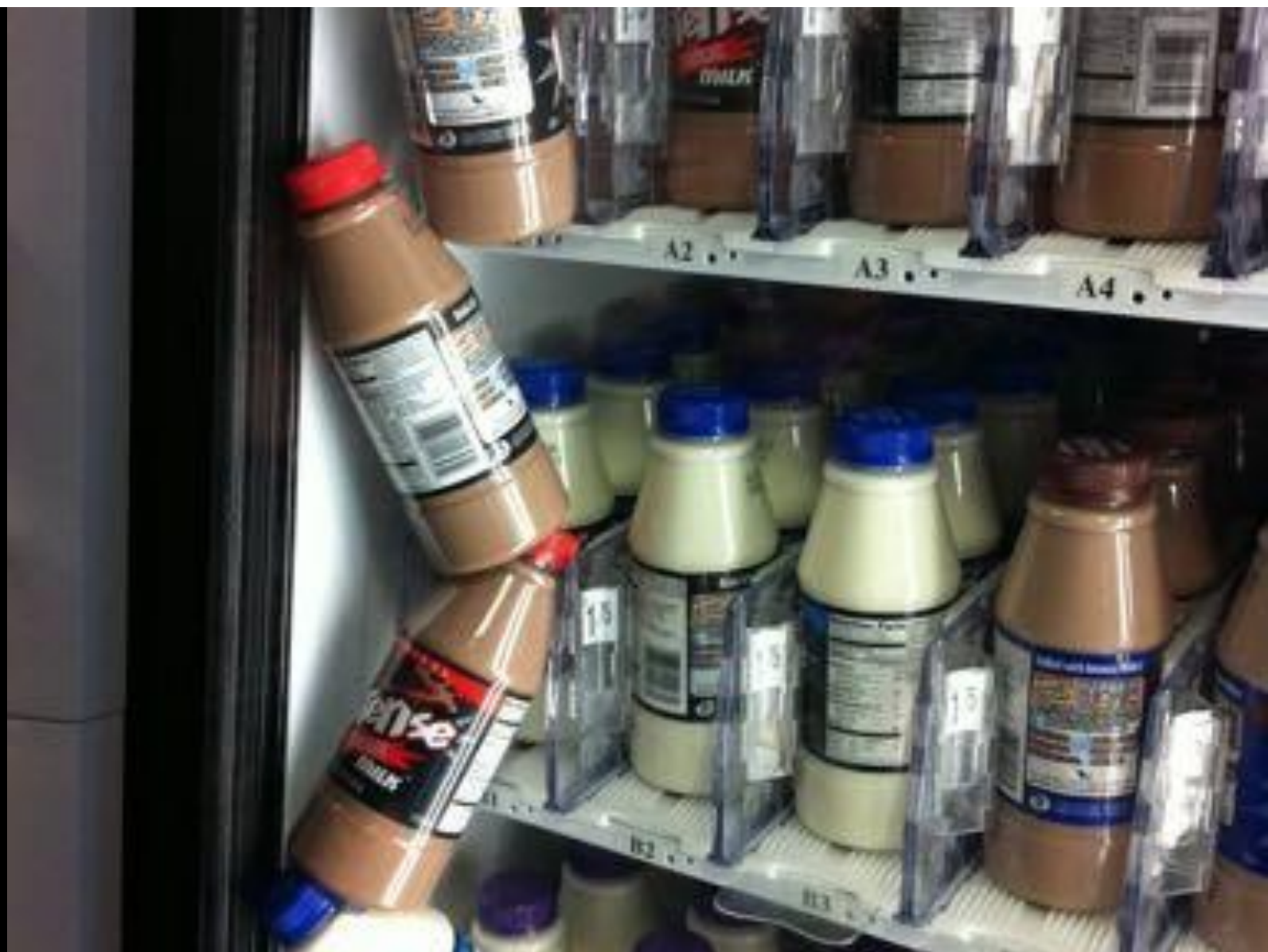


VEND • CAFE

BURGER'S



4)
Columet





Trying the Understand what Users want!

- Do users want anything else with their soda?
- What methods of payment do users prefer?
- What do users want when a machine fails?



Why do a User Needs Assessment?

- To understand **why** people **want** to use a product or service?
- To learn **how** people actually **use** or don't use a product or service
- To see **what** people **like or dislike** about a product or service
- To hear **what else** people might **want** from a product or service

When to do a User Needs Assessment?

- **Before any design phase**
 - *Before designing something new*
 - *Before a re-design*
- **Anytime questions or problems arise**
 - *Why would users want to use this service and*
 - *What do they think about each time they use it?*
 - *Why don't customers go beyond the home page of a particular website?*
 - *What frustrates users of a given project?*
 - *How could we improve the client's experience?*

“Needs” are not always urgent

- **Origin of Need Assessment**
 - 1960
 - *Roger Kaufman, an educational technology researcher*
- **In UX, “needs” = needs, wants, preferences, quirks**
 - *Often focused on a product or a service*
 - *Pros and Cons to taking a broader view*

Qualitative Research

- **Initial Focus:**
 - *Interviews*
 - *Observations*
 - *Affinity Walls*
- **Strengths of Qualitative Research**
 - *Uncovering ideas and concepts In-depth*
 - *Broad range of use cases*
 - *Hypothesis generation about what users really want*
 - *Causal explanations*

Other methodologies

- Work Models
- Personas
- Market research
- Surveys and Questionnaires
- Log Analysis
- Document Analysis

Design a new GPS version?

Let's suppose that the general manager of a GPS device product comes to you and says, can you help us design a new version?



Sales Data? Survey? Usage Stats?

- Quantitative research helps to understand...
 - *Overall trends*
 - *Existing product or service*
 - *Issues you can anticipate*
- Quantitative research is *less useful* for...
 - *Detailed understanding*
 - *New possibilities*
 - *Unanticipated issues*



More Interesting Questions...

- Detailed Understanding

How do users input a destination?

Do they make multiple detours through different methodologies because they get stuck?

- Are they really frustrated at any phase?

- New feature

Quizzes to help people memorise routes

- Unanticipated Issue

- *Might families want 'most fun' route?*



Qualitative Research

- Characterized by...
 - *Non-numerical data*
 - *Direct interaction with people and contexts*
 - *Richness and detail*
- Strengths of qualitative research
 - *In-depth*
 - *Broad range*
 - *Hypothesis generation*
 - *Causal explanations*



Qualitative Research Methodologies

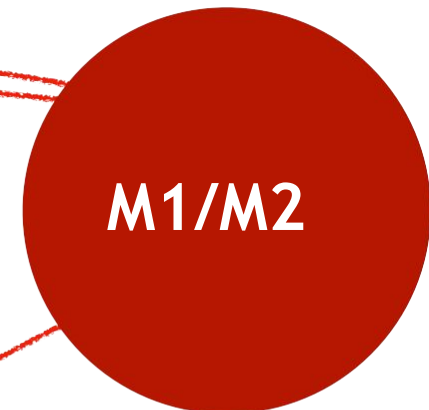
- Data gathering:

- *Interviews*
- *Field observations*
- *Participant observations*
- *Focus groups*
- *Etc.*

- Data analysis:

- *Coding*
- *Content analysis*
- *Thematic analysis*
- *Recursive abstraction*

] Via Affinity
Diagrams



Qualitative Research Caveats

- Qualitative research is not the best for questions such as...
 - *What proportion of our users do X?*
 - *Would usage increase if we did Y and instead of Z?*
 - *Does program V cause effect W?*
- Weaknesses of qualitative research:
 - *Providing numerical data or analysis*
 - *Confirming hunches, guesses, intuitions*
 - *Making close comparisons*





Qualitative research is useful in a wide range of contexts beyond user needs assessments.



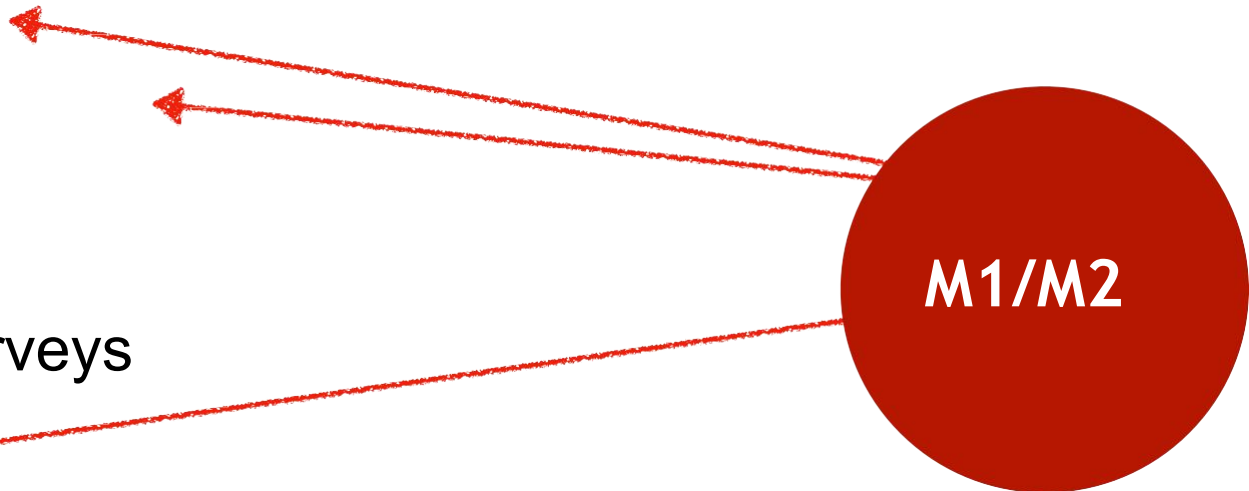
Qualitative research is useful in a wide range of contexts beyond user needs

Semi-Structured Interviews



Types of Interviews

- Group vs. **individual**
- Remote vs. **in-person**
- Degree of structure
 - *Structured*
 - Questionnaires/Surveys
 - ***Semi-structured***
 - *Unstructured*



Semi-Structured Interviews

- **Balance between...**
 - *Researcher interests and focus*
 - *Open, free-form discussion*
- **Script, or interview protocol**
 - *A list of questions that you use as a guideline for the interview*
 - *Won't actually follow them strictly.*
- **Conversational style**
- **Expect to go off-script!**

Interviews for Needs Assessment

- **Who**
 - *User / potential user of product or service*
 - *“Typical” member of group you are interested in*
- **Where**
 - *Where participant normally uses product or service (or might do so)*
 - *Where participant feels comfortable*
- **When**
 - *Scheduled in advance*
 - *Mutually convenient time*
 - *When participant normally uses product or service (or might do so)*

Components of an Interview Protocol

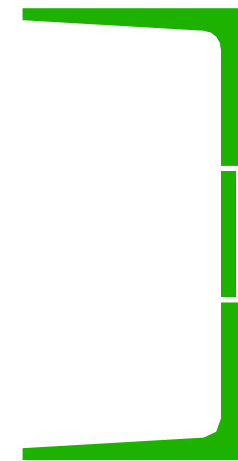




How to improve the situation at airport security?

Components of an Interview Protocol

- Overarching question
- Introduction
- Main interview
 - *Background / warm-up questions*
 - *Core questions*
- Observation
 - *Think-aloud protocol*
 - *Observation questions*
- Conclusion



One set for
every type
of interview
participant

Overarching Question

- This is the *big question* you are trying to answer.
- Keep it in mind *throughout* protocol development and actual interview.
- You *will not* necessarily ask the interview participant this question *directly*.



Introduction

- Script for you to explain process to interview participant:
 - *Self-introduction*
 - *Goals of interview*
 - *Duration*
 - *Overview of components*
 - *Confidentiality*
 - *Voluntary nature*
 - *Permissions*

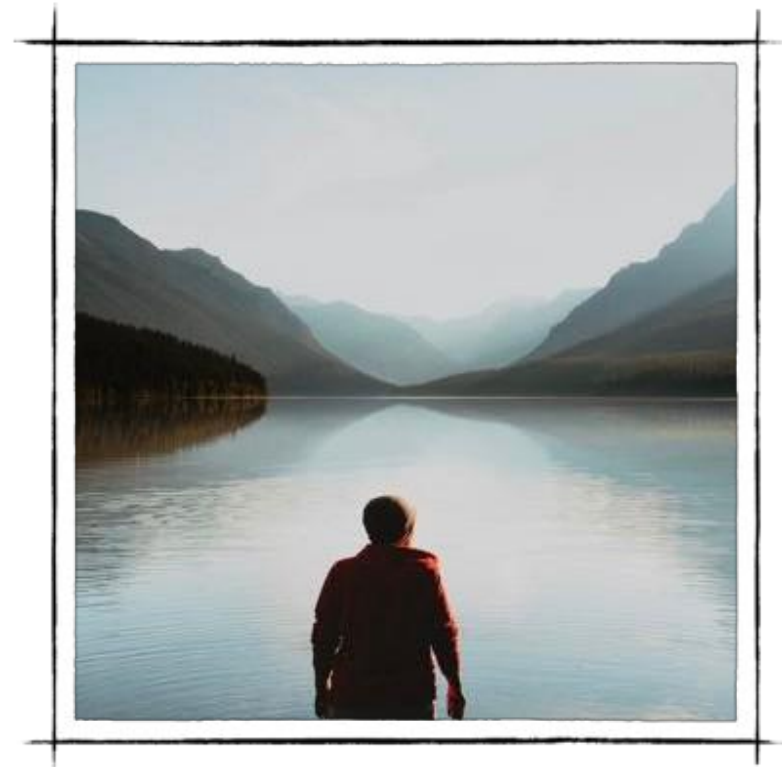
Main Interview



- Background / warm-up questions
 - . *Short Simple*
 - . *Closed-ended*
- Core questions
 - . *Open-ended*
 - . *Conversational*
 - . *Focused on overarching question*

Observation

- **Think-aloud protocol**
 - *Encourage participant to verbally explain what they are thinking and doing.*
 - *Awkward at first; gets easier.*
- **Observation questions**
 - *Prompts to perform tasks*
 - *Questions you can anticipate*
 - *Focused on overarching question*



Conclusion

- **Script for the end of the interview and observation:**
 - *“Anything to add?”*
 - *Thank you*
 - *Follow-up plans*
 - Additional questions
 - Report back
 - *Your contact information*
 - *“Any questions?”*
 - Thank you, again!

Additional Comments

- Overarching question
- Introduction
- Main interview
 - *Background / warm-up questions*
 - *Core questions*
- Observation
 - *Think-aloud protocol*
 - *Observation questions*
- Conclusion
- Guidelines only - no hard rules
- Most important:
 - *Interview ethics*
- Second most important:
 - *Answering your overarching question*
- Variations
 - *Multiple overarching questions*
 - *Interleaving interview and observation*
 - *No observation*
 - *In-situ observation, with no think-aloud protocol*

Good Practices for Interview Protocols





"Just one more thing..."

Detective Columbo
(played by Peter Falk)

Combining the Science and Art of Interviewing

Interviewing as Science and Art

- Preparation

- *Interview participant background*
- *Topic background*
- *Interview protocol*
- *Overarching question*

- Improvisation

- *Rapport with interview participant*
- *Attention to the unexpected*
- *Unscripted questions*
- *Focus on overarching question*

Good vs Not so Good Questions!

- **Good Questions**

- **Open-ended**

- *“Why did you...?”*
- *“How did you go about...?”*
- *“Can you tell me about...?”*
- *“Why do you think that is?”*
- *“How did you handle that situation?”*

- **Not So Good Questions**

- **Closed-ended**

- *“Do you like X?”*
- *“How many times a week do you...?”*
- *“What is your favorite...?”*
- Sometimes, closed-ended questions are appropriate, but they shouldn't dominate the protocol.

Good vs Not so Good Questions!

- Good Questions

- Specific, concrete

- *“Tell me about the last time that you did...”*
- *“Can you think of a specific instance when...?”*

- Not so Good Questions

- Abstract, generalized

- *“What do you normally do when...?”*
- *“What is the typical way in which...?”*

- These aren't bad questions, but you want to be sure that you've asked for concrete details, also.

Good vs Not so Good Questions!

- Good Questions

- Non-leading, judgment-free

- *“What option do you prefer?”*
- *“What do you think was the reason for that?”*
- *“How did you try to address the problem?”*

- Not so Good Questions

- Leading, judgmental

- *“So, you prefer X over Y?”*
- *“Was it because of a bad process that the task was difficult?”*
- *“Why didn’t you use feature Z?”*

Organization and Flow

- Cluster related questions together.
- Within each cluster, arrange them in a meaningful order:
 - *Chronological*
 - *Narrative*
 - *Broad to narrow, or narrow to broad*
 - *Etc.*
- Arrange clusters in a meaningful order.
- Save any sensitive questions for the end.

Follow-Up Questions

- Under each core question, add follow-up questions.
 - *Use bulleted lists.*
- Follow-up question types:
 - *Generic:*
 - “Why?” “How did that happen?” “Tell me more.”
 - *Prompts for specific issues:*
 - “Did you think about X?” “Or Y?”
 - *Repetition:*
 - “Could you tell me about another time when...?”
 - *Etc.*
- Think through the order of these, as well.
- Again, you will probably not ask all of the follow-up questions.

Final Thoughts on Interview Protocol

- **Goals of protocol:**
 - *Preparation*
 - *Guide during interview*
- **Memorize the overall flow of the protocol.**
- **Remember - you will not read protocol questions one after the other.**
- **See sample and guidelines**
 - see GCR, interview template is provided under assessment milestone 1

Tips for Interviewing



Interviewing as Science and Art

- Preparation
 - *Interview participant background*
 - *Topic background*
 - *Interview protocol*
 - *Overarching question*
- Improvisation
 - *Rapport with interview participant*
 - *Attention to the unexpected*
 - *Unscripted questions*
 - *Focus on overarching question*

Rapport with Interview Participant

- Make them feel comfortable.
- Do not jump right in.
- Start with small talk.
 - *Try -----*
 - Finding something in
 - common Complimenting
 - Discussing the weather
- Adopt a learning mindset.
- They are the expert.
- Observe
- Adapt
- Listen
- Listen
- Listen!

Follow Interesting Threads

- **Advanced preparation pays off.**
 - *Let go, but keep overarching question in mind. Use protocol as back-up.*
- **Ask follow-up questions until satisfied.**
- **Listen for unexpected but relevant things.**
 - *Improvise unscripted follow-up questions.*
- **Be curious!**

Note Taking

- Try using paper and pen/pencil.
 - *Devices can create a mental barrier.*
 - *Print out interview protocol with blank space under core questions.*
- Write down...
 - *Key points, in short phrases*
 - *Things that audio recording would not capture: impressions, facial expressions, physical context...*
 - *Follow-up questions*
 - *Etc.*
- Important, even if audio recording.
- No “right” way to take notes. Develop your own style.





How do you get to MM Alam Road?

Practice, Practice, Practice

- Maintain conversational tone.
- Keep overarching question in mind.
- Be curious!
- *See samples for more tips.*



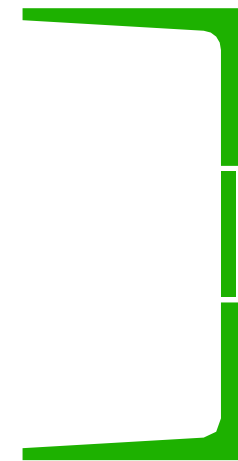
Observing Users

Observing Users



Components of an Interview Protocol

- Overarching question Introduction
- Main interview
 - *Background / warm-up questions*
 - *Core questions*
- Observation
 - *Think-aloud protocol*
 - *Observation questions*
- Conclusion



One set for every type of interview participant

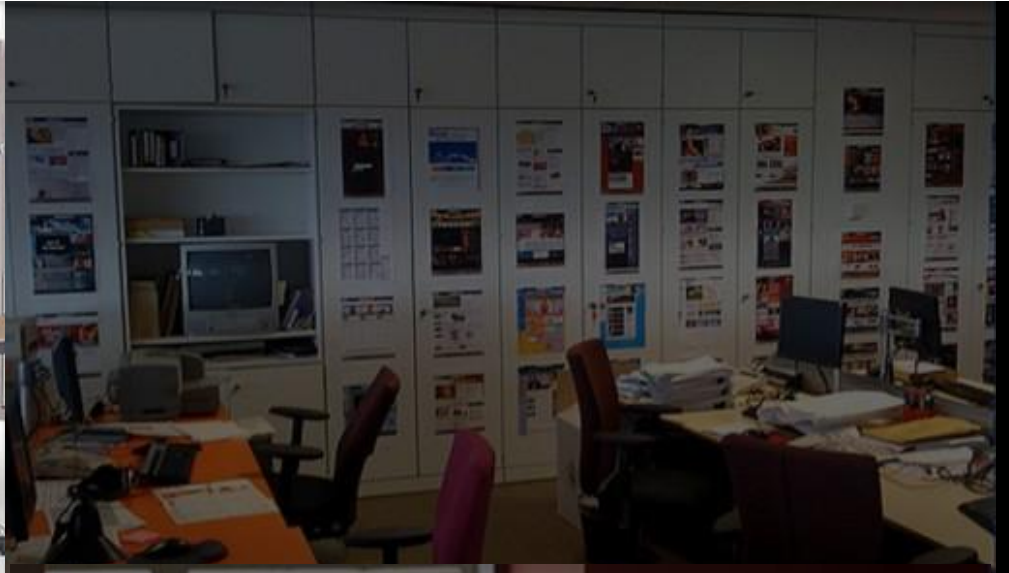
Observation

REVIEW

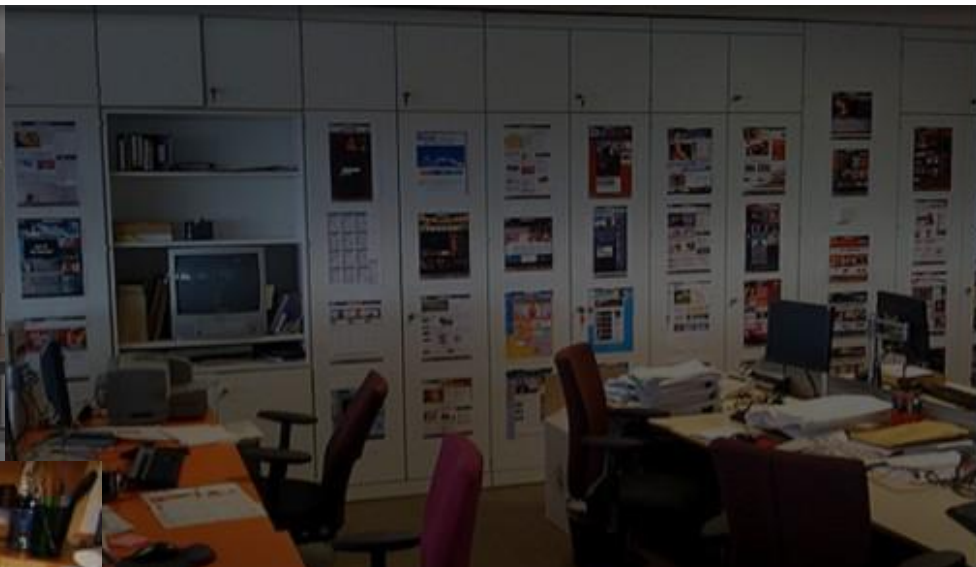
- **Think-aloud protocol**
 - *Encourage participant to verbally explain what they are thinking and doing.*
 - *Awkward at first; gets easier.*
- **Observation questions**
 - *Prompts to perform tasks*
 - *Questions you can anticipate*
 - *Focused on overarching question*



Observation starts even before the Interview









Observation Tips [1/2]



- Ask participant to perform relevant tasks.
- Think-aloud protocol
 - *Ask participant to speak aloud and tell you what they are doing and thinking.*
 - It will be awkward at first, but encourage them if needed, by asking lots of quick questions:*
 - “What are you doing now?”
 - “What are you thinking when you do that?”
 - Etc.

Observation Tips [2/2]

- Take good notes.
- Pay special attention to things they do that are not part of the main task:
 - *Pauses*
 - *Missteps / multiple attempts*
 - *Intentional detours Etc.*
- Ask questions, if you don't understand the motivation for an action.

Otherwise, Similar to Interviews

- Maintain conversational tone.
- Keep overarching question in mind.
- Be curious!
- Practice, practice, practice.



Extracting Data from the Interviews

Extracting Data from Interviews



Note Taking

REVIEW

- Try using paper and pen/pencil.
 - *Devices can create a mental barrier.*
 - *Print out interview protocol with blank space under core questions.*
- Write down...
 - *Key points, in short phrases*
 - *Things that audio recording would not capture: impressions, facial expressions, physical context...*
 - *Follow-up questions*
 - *Etc.*
- Important, even if audio recording.
- No “right” way to take notes. Develop your own style.

Data Extraction



- **Raw data**
 - *Notes*
 - *Audio Recording*
- **Make some sense of the Raw Data**
 - *Process called data extraction*

From Notes to Qualitative Data

- As soon as possible after an interview, review notes and audio recordings and write down affinity notes.
- An affinity note is...
 - *a sticky note on which you write*
 - a participant code (type and number: C01, C13, G04, M02, etc.)
 - a statement or question
 - that comes from your notes or audio recordings
 - which is probably relevant to the needs assessment.

Affinity Note Guidelines [1/2]

- Include...
 - *Factual statements – complete sentences*
 - *Participant quotations (in quotation marks)*
 - *Observations, as statements*
 - *Interpretations of the above*
 - *Questions*
- **Tips:**
 - *Make each note understandable on its own.*
 - *Be concrete but concise.*
 - *Aim for about one affinity note per minute of interview.*

Affinity Note Guidelines [2/2]

- **Interpretations:**
 - *Repeating themes*
 - *Contradictions or conflicts*
 - *Key findings*
 - *Root issues*
 - *Anything else that might be relevant*
- **Good quotes:**
 - *Unexpected*
 - *Representative*

Assignment - M1/ M2

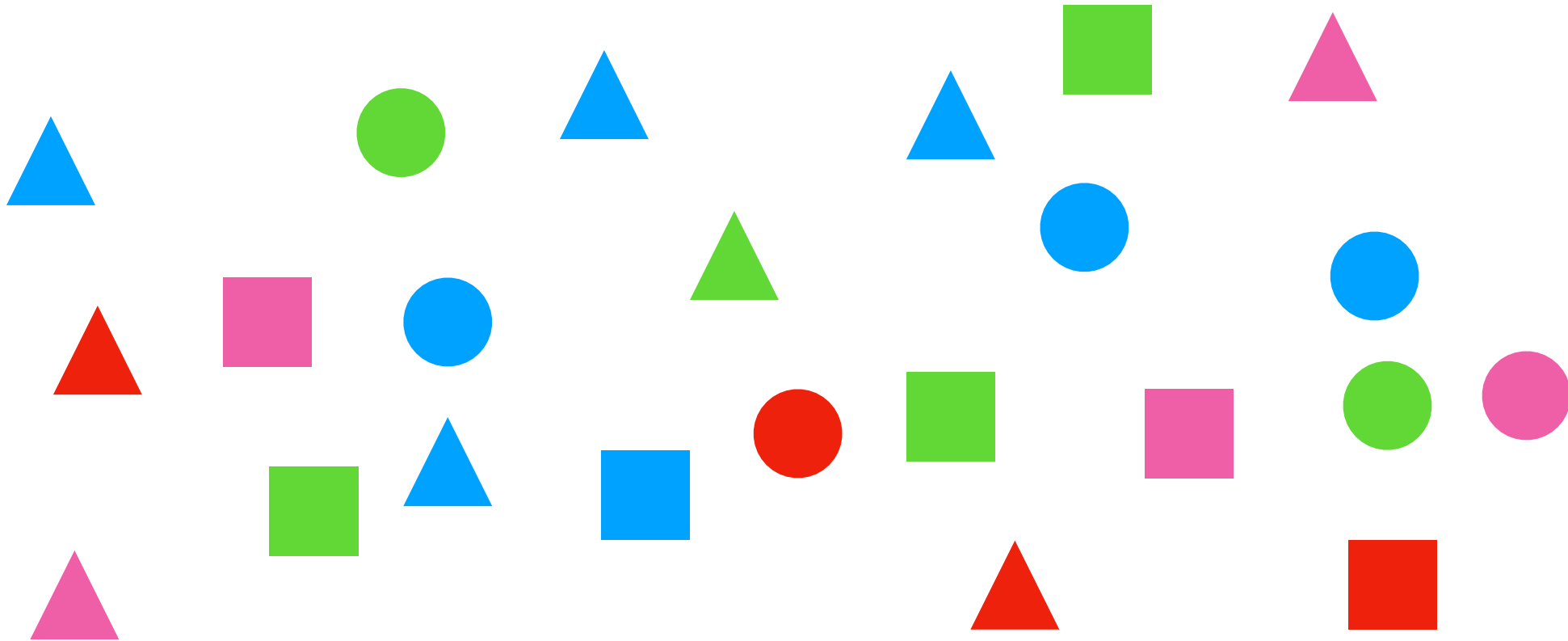
*Interviews, Observations
and Affinity Notes*



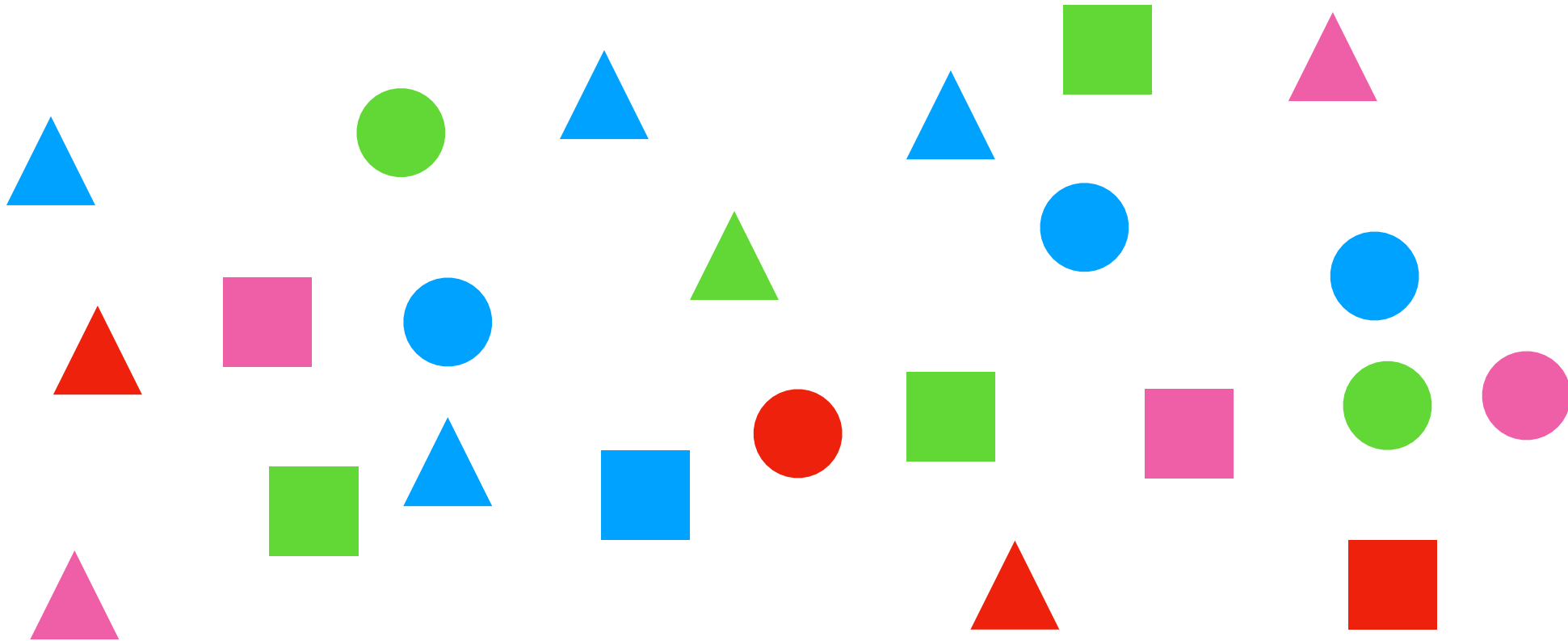
Analysing Qualitative Data



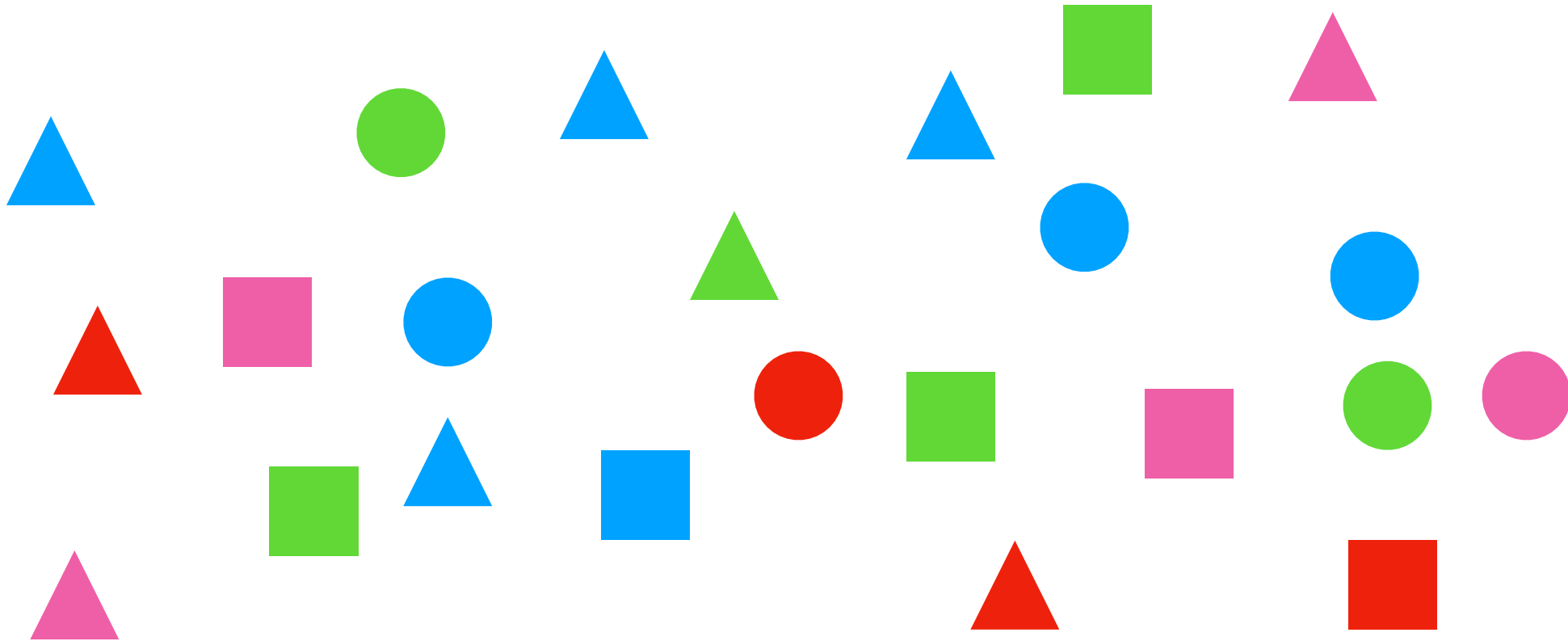
Can you memorise all the items?



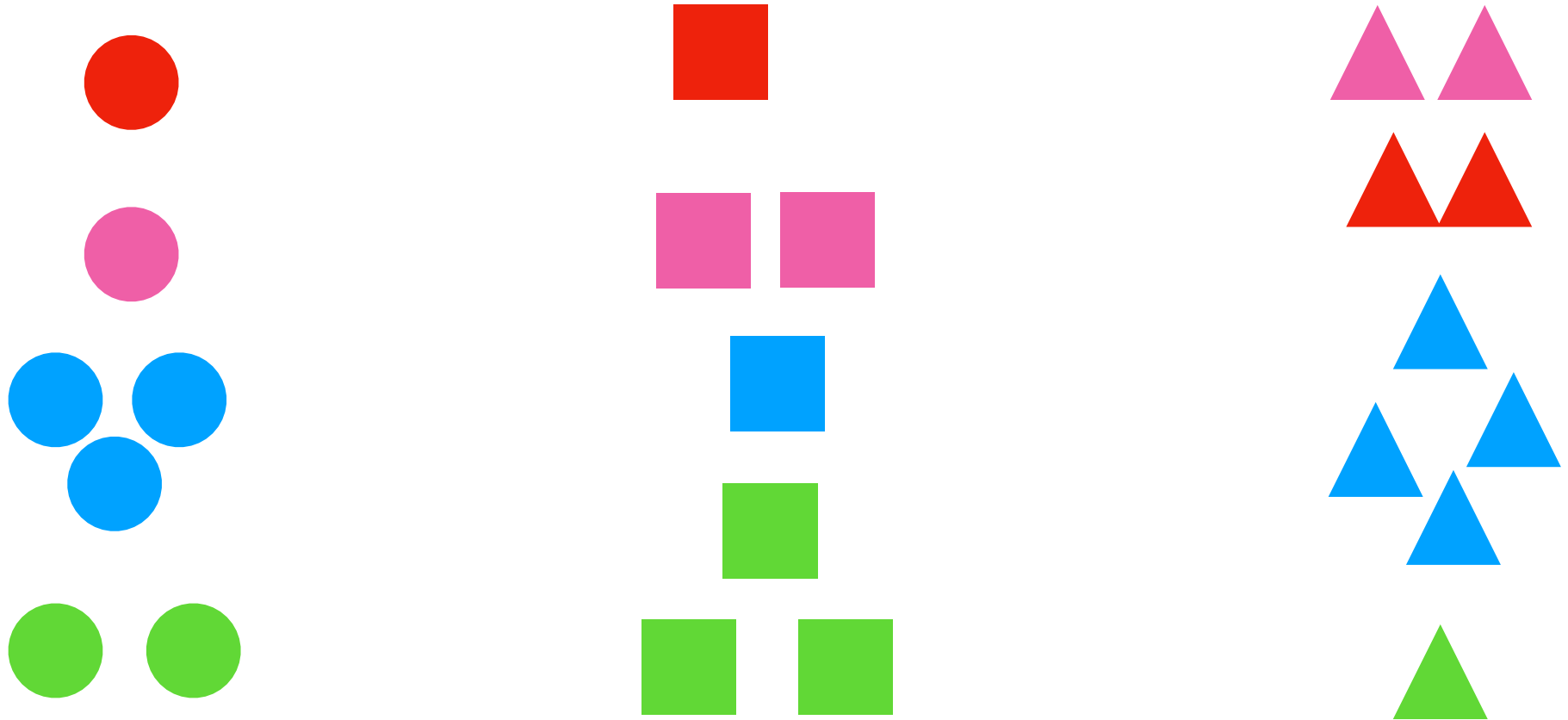
Cluster



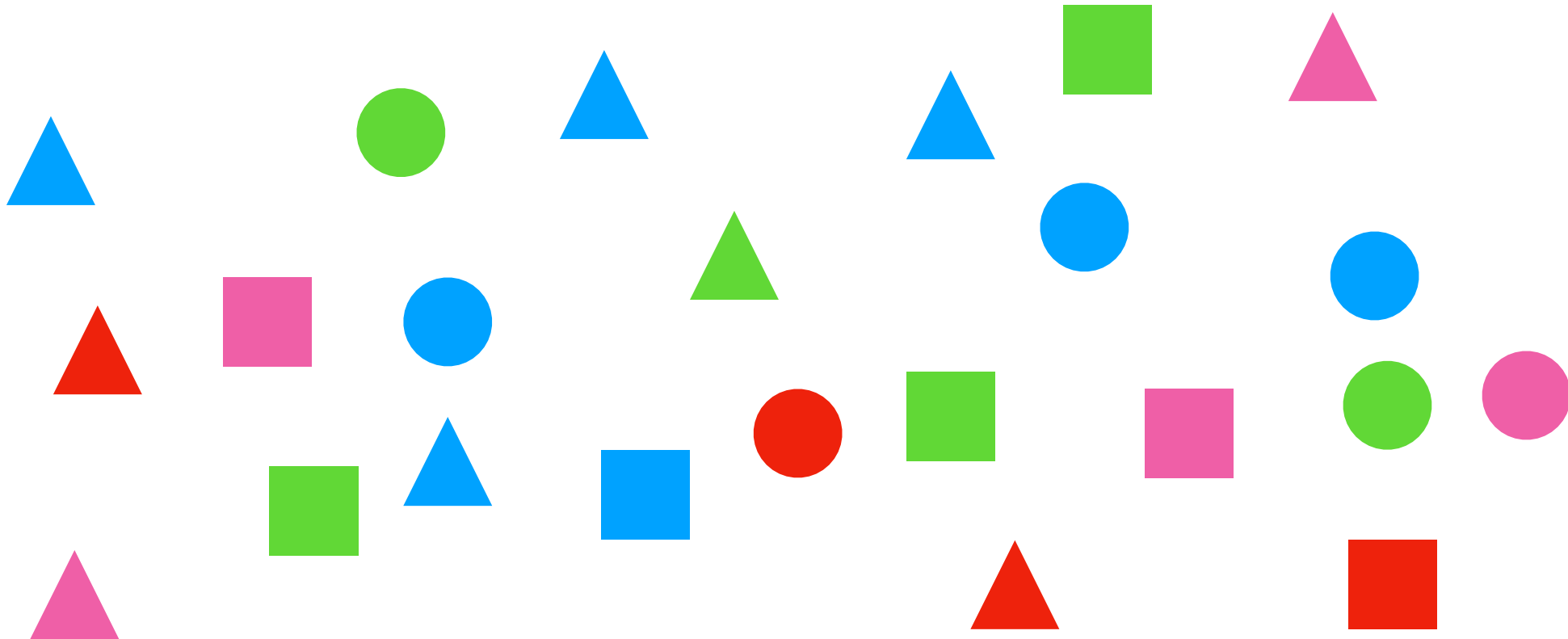
Cluster and Sort



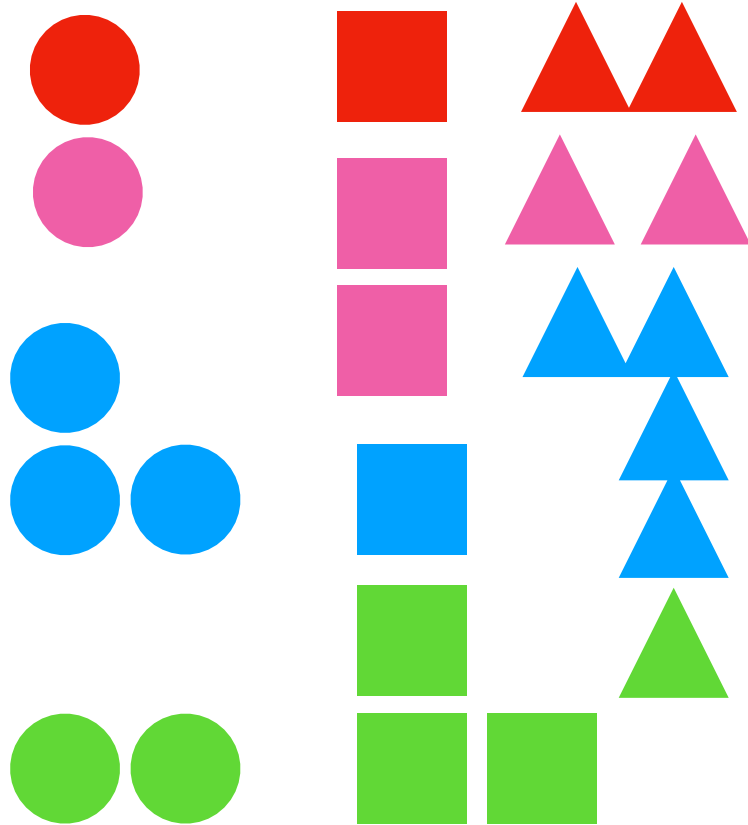
Sort & Notice Patterns



Easier than...



The Secret to Intelligence!



- The world is full of information.
 - *You can't remember it all.*
- By clustering, sorting, finding patterns, and relating things to each other, you can condense information.
- Condensing is understanding.
- Understanding is intelligence!



Analyse your data with an affinity wall...

Affinity Walls





Affinity Walls - way to do Qualitative Data Analysis



Affinity Wall Demo



Affinity Wall Demo

Affinity Wall Guidelines

- **At first, make clusters quickly.**
 - *Don't get too attached to clusters.*
 - *Move notes around as you see new patterns.*
- **Re-consider clusters.**
 - *Aim for clusters of 3-7 notes.*
 - *Break up if more than 7 notes.*
 - *Merge if less than 3 notes.*
- **Not a hard rule!**

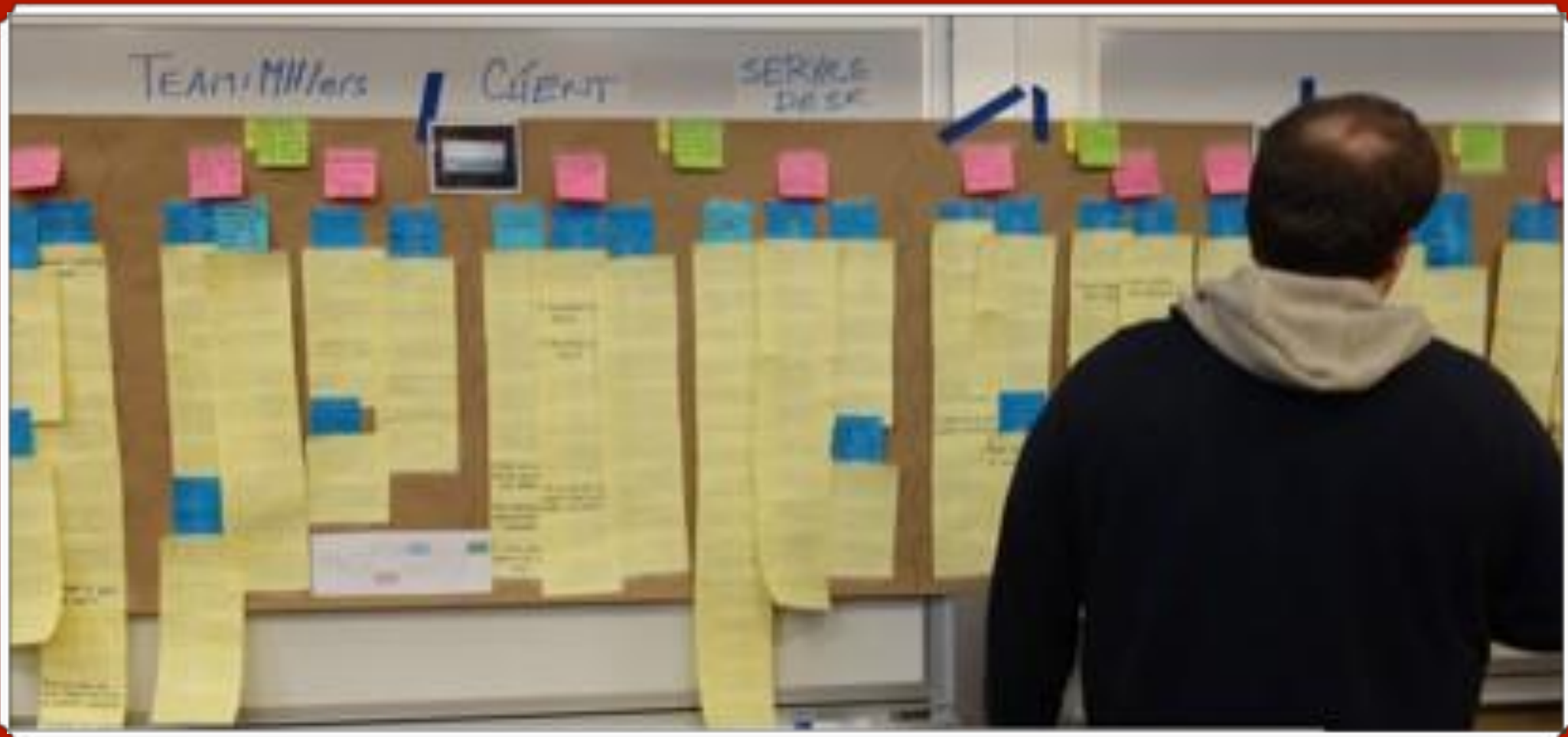
Affinity Wall Guidelines

- Writing the next-level notes is the crux:
 - *Summarize the knowledge in the cluster.*
 - *Summary should be a full sentence and make sense on its own.*
- Balance abstraction and precision.
- Think!
- Stop creating levels when summaries become uninformative.



Affinity Wall Assignment

as part of
M2



Good Luck and Have fun with Affinity Walls!

Variations on User Needs Assessments



Qualitative Research

REVIEW

- **Methods in this course:**
 - *Interviews*
 - *Observations*
 - *Affinity Walls*
- **Strengths of Qualitative Research**
 - *Uncovering ideas and concepts In-depth*
 - *Broad range of use cases*
 - *Hypothesis generation about what users really want*
 - *Causal explanations*

Other methodologies

REVIEW

- **Work Models**
- **Personas**
- **Market research**
- **Surveys and Questionnaires**
- **Log Analysis**
- **Document Analysis**

Qualitative Research

REVIEW

- **Initial Focus:**
 - *Interviews*
 - *Observations*
 - *Affinity Walls*
- **Strengths of Qualitative Research**
 - *Uncovering ideas and concepts In-depth*
 - *Broad range of use cases*
 - *Hypothesis generation about what users really want*
 - *Causal explanations*

Related Concepts [1/2]

- The kind of qualitative research you learned is sometimes called...

*Ethnographic
research*

Contextual inquiry

- Affinity walls are also called...

*Affinity
diagrams The
KJ Method*

- Affinity walls are one way to do...

- *Qualitative analysis*

-



Related Concepts [2/2]



- User needs assessments are a part of...
 - *User-centered design*
 - *Human-centered design*
 - *Ethnographic design*
 - *Empathic design*
 - *Contextual design*
 - *UX design*
 - *UX research*

Gerry
Gaffney

Contextual Enquiry (site visits)

- 1) IDENTIFYING THE USERS
- 2) SCHEDULING AND CONDUCTING VISITS
- 3) ANALYSING THE DATA



\$

DATE	ACTIVITY
2-3	PLANNING
3	VISITS
2-3	ANALYSIS



**2 VISITS
PER DAY**

LET USERS KNOW
WHAT TO EXPECT



Contextual Inquiry



Wide Applicability

Conclusion





No Rocket Science - Rules are Simple!

Involves a lot of practice so that you can really get into the groove of how to put these ideas into practice.